

Gregory Ryan Seaton

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Summary

Machine learning and AI researcher focused on optimizing workflows in modern ML pipelines. Experienced in model development and designing efficient, scalable processes that improve deployment outcomes, with an emphasis data-driven decision optimization.

Education

Washington University in St Louis: Bachelor of Science, Data Science

2023 - May 2027

- **GPA:** 3.88
- **Achievements:** Danforth Scholar, Synchrony Banking Scholar, Bauer Leadership Program Inductee, Delta Sigma Pi Brother
- **Coursework:** Data Science Statistics, Data Structures & Algorithms, Linear Algebra, Decision and Estimation Theory for Stochastic Processes, Optimization, Linear Statistical Models, Managerial Economics, GIS for Spatial Science, Managerial Statistics
- **Skills:** Python, TensorFlow, PyTorch, SQL, R, Excel, MongoDB, Tableau, ArcGIS, Bash, Java, React, HTML, MATLAB, GitHub, Dynamic Programming, Integer Programming, Convex Optimization, Linear Optimization, Stochastic Dynamical Systems

Work Experience

National Science Foundation: IIT Madras & ePlane Company

May 2025 – August 2025

International ML Research Fellow

Chennai, Tamil Nadu, India

- Designed an ML-driven modeling framework to learn and predict operational patterns in autonomous air-taxi networks, using data-centric pipelines to capture dispatch behavior, battery usage, and vertiport capacity constraints across Indian urban regions.
- Built a stochastic demand-modeling pipeline using Poisson processes and temporal feature engineering to simulate 400,000 daily trips.
- Developed a TensorFlow-based geospatial clustering model on taxi GPS data to identify 11 high-demand hubs and translated cluster outputs into downstream-compatible ML features for deployment planning and user-centric business experimentation.
- Delivered a 30%+ decrease in average wait times through a decision-optimization engine that used a global scoring model to evaluate feasible aircraft-request pairings, improve assignment quality, and maintaining system-wide fleet availability.

University of Utah

May 2024 - Present

Systems Engineering Research Intern

Salt Lake City, Utah

- Formalized an AWS autonomous-vehicle security testbed that increased experimental reproducibility by 45% (measured by reduction in run-to-runs variance) and improved the reliability of cyber-physical system evaluations.
- Designed and conducted 100+ adversarial ML experiments, demonstrating computer vision classification-confidence drops of 40–60% under physical-world perturbations, and identifying critical perception vulnerabilities.

Beats By Dre

Jul 2024 - Sep 2024

Consumer Insight Analyst Extern

Remote

- Executed large-scale sentiment analysis on 1,000 Amazon reviews using NLP methods to evaluate consumer satisfaction for Beats and competing audio brands, producing comparative sentiment and feature-level insights.
- Built an automated web-scraper and a custom text-processing pipeline using regex normalization, tokenization, and VADER to compute polarity scores across 10 different product lines

Projects

Twitch.Tv Analytics Insights

Jul 2025

- Built a full-stack analytics tool using the Twitch API to monitor VODs, clips, and stream metrics across creator accounts.
- Delivered personalized recommendations to help streamers optimize game selection, stream timing, and content strategy.

Geospatial Vessel Routing Optimization

Apr 2025

- Constructed a multi-factor geospatial analytics model in ArcGIS Pro to identify optimal vessel routes across the Gulf of Mexico, integrating ocean-current vectors, historical hurricane tracks, and movement patterns from 300K+ AIS vessel records.
- Engineered a scoring framework that weighted environmental and behavioral features to produce least-cost navigation routes.

Parkinson's Disease Detection Using Voice-Based Machine Learning Models

Apr 2025

- Implemented TensorFlow CNN to classify Parkinson's disease from biomedical voice features, achieving 94% diagnostic accuracy.
- Replicated full ML pipelines including feature extraction, normalization, hyperparameter tuning, and model-comparison workflows to evaluate performance across classical and deep-learning architectures.

Leadership

Data Science Network

May 2024 - Present

Founder, President

- Founded a 185+ member student organization focused on data science career exploration and interdisciplinary learning

Washington University in St. Louis Admissions

Jan 2025 -Present

Bear Ambassador

- Deliver engaging and informative campus tours to over 400+ prospective students and families

Data Structures and Algorithms Curricula Team

Jul 2024 - Jan 2025

Teaching Assistant

- Conducted weekly sessions with 15 students to practice implementing data structures throughout the academic year
- Evaluated over 520 assignments and exams while offering constructive feedback to help students improve their performance

Interests

International Solo Traveling, Pokémon, Greenery, Soccer, Hiking, Meditation, Reading, Board Games